

Section 1-1: Characteristics of Cloud Computing

- **Cloud Computing Certifications:**

There are **two main certifications** for cloud computing:

- **Cloud Essentials+** is for people who are not IT professionals or for IT workers who want to understand both business and technical sides of cloud computing.
- **Cloud+ (CV0-003)** is more technical and **and skills**.

- **Cloud+ builds on other CompTIA certifications:**

Cloud+ certification is based on the knowledge from:

- **CompTIA A+** (basic computer skills)
- **CompTIA Network+** (networking knowledge)
- **CompTIA Security+** (security concepts)

- **Meaning of Cloud+:**

Cloud+ is an **infrastructure-specialty certification**, meaning it focuses on the technical structure of cloud systems.

It connects the basic IT skills (A+, Network+, Security+) with advanced cloud topics such as infrastructure and security in cloud environments.

CompTIA Cloud+ Certification

- The **Cloud+ certification** is *vendor-neutral*, which means it is not limited to one specific company or cloud platform.
- People study for this certification for several reasons:
 - To prepare for a **new job or promotion**
 - To **build a strong basic understanding** of cloud computing before learning vendor-specific systems.
 - To **add value** to other technical certifications.
 - To gain a **bigger understanding** of cloud technologies

Other Cloud Certifications

- **Vendor-neutral** certifications:
 - **CompTIA Cloud+** – teaches general cloud computing knowledge.
 - **CCSP (Certified Cloud Security Professional)** – focuses on cloud security skills.
- **Vendor-specific** certifications:
 - **AWS (Amazon Web Services)**: Offers many certifications for different cloud roles.
 - **Microsoft Azure**: Provides several certifications in cloud technologies.
 - **GCP (Google Cloud Platform)**: Offers certifications for cloud operations and development.
 - **VMware**: Offers *Cloud Management and Automation* and *Certified Professional* programs, which connect with public cloud platforms.

What is Cloud Computing?

- Cloud computing has the following five essential characteristics:



- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

Additional benefits include:

- Self-patching/self-healing infrastructure:** The system can automatically fix problems and update itself.
- Adaptive, intelligent security:** Security systems learn and adjust to new threats.
- Cross-platform:** Cloud services can work on different operating systems and devices.

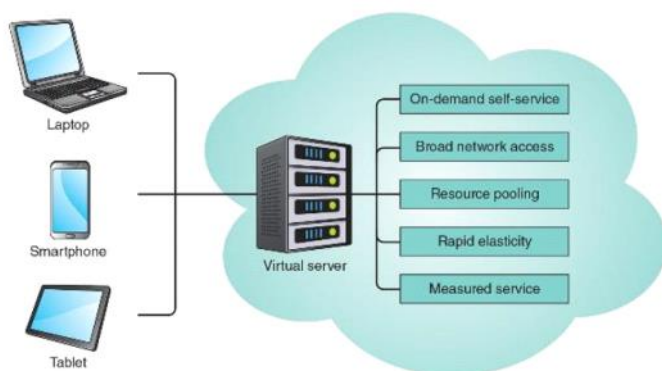


Figure 1-2 Characteristics of cloud computing



Organizations **move to cloud computing** for several important reasons:

- Lease expiration:** When the lease for a physical data center ends, (lease = kiralama)
- Service optimization:** Moving to the cloud **helps improve services** and **makes companies more competitive**
- Better financial performance:** Cloud computing **helps reduce costs** and **increase profits**.

Most organizations do not move to the cloud all at once.

Instead, they **adopt cloud services step by step (in phases)** — starting small, testing, and then expanding as they gain more confidence and experience.



What Do I Need to Know? (1 of 2)

- Security **should be built into all cloud resources**.
- Understand the **data moved to the cloud**.
- Know **business goals** and workflows.
- Learn **basic software development and infrastructure skills**.
- Be aware of **security risks** and best practices.
- Use automation tools** for efficiency.
- Keep learning **as technology changes fast**.

What Do I Need to Know? (2 of 2)

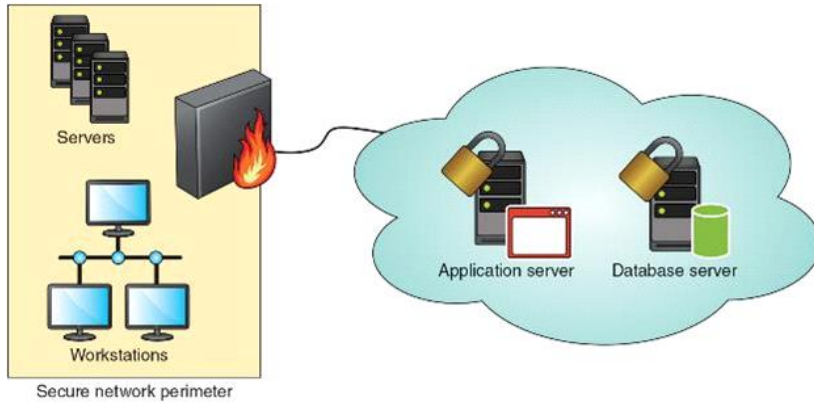


Figure 1-3 Cloud security can't rely on a secure perimeter

! This image shows that **cloud security cannot depend only on a secure perimeter** (like a firewall).

In traditional IT systems, security is focused on protecting the **network boundary** — everything inside is trusted, and everything outside is not.

But in **cloud computing**, data and applications move across the internet and between different systems. So, instead of protecting just the network, **each cloud resource (like application or database servers) must have its own security** — such as encryption, authentication, and access control.

Cloud Deployment Models

- Cloud computing gives **easiness** but also means **less control** over hardware.
- Businesses can use **another company's hardware** to **store data and run applications**.
- However, they must **trust that company's systems** for performance and security.
- This dependence can create **different security risks**, depending on which cloud model is used.

- Public Cloud

- Public cloud services use the **provider's (CSP's)** hardware, not the customer's.
- The **CSP manages and secures** the hardware — customers can't access it directly.
- Customers depend on the **CSP's security measures** to protect their data. !
- Before using a provider, customers should check the **CSP's certifications, audits, and SLA (Service Level Agreement)** for security details.

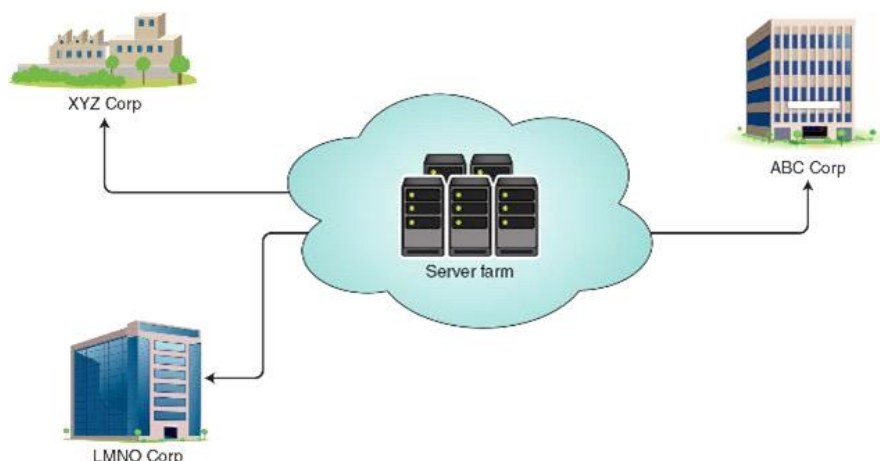


Figure 1-4 A public cloud is hosted on hardware shared by multiple customers

CSP = Cloud Service Provider = company that offers cloud-based services
ex = Azure , AWS , GCP

- Private Cloud

- A **private cloud** is used **only by one organization**, not shared with others.
- The hardware can be **in the company's own data center** or **in a CSP's data center**.
- Only the organization can access it, which means **better security**.
- It is different from a normal data center because it uses **APIs** and may include virtualization.
- Security is handled in a **more traditional, internal way**.

Private cloud = **Exclusive use + stronger control + higher security**.

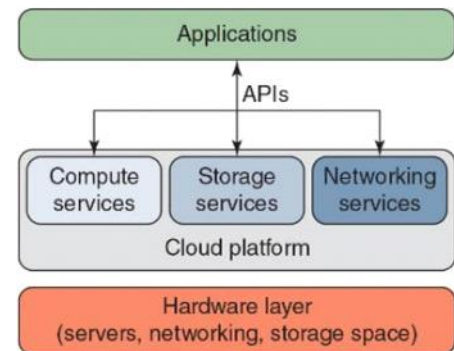


Figure 1-5 Cloud services and underlying hardware communicate with APIs

- Hybrid Cloud

- A **hybrid cloud** combines **public and private cloud** systems, or mixes **cloud and traditional local (on-premises)** services.
- Some companies move to the cloud **partially**, because not all data or apps fit the cloud.
- It is the **most common cloud model**, since many organizations aren't ready to move everything to the cloud.
- **Security** in hybrid cloud includes measures from **both public and private clouds**.

Hybrid cloud = **Mix of public and private cloud** → flexible, but needs careful **security management**.

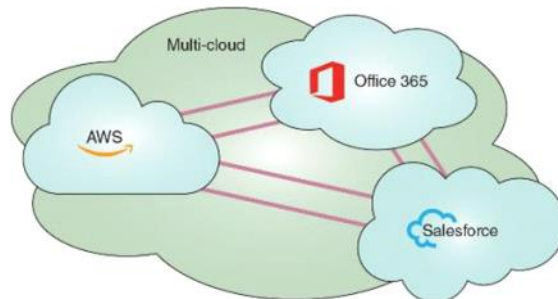


Figure 1-7 A multi-cloud intertwines cloud resources from multiple cloud providers

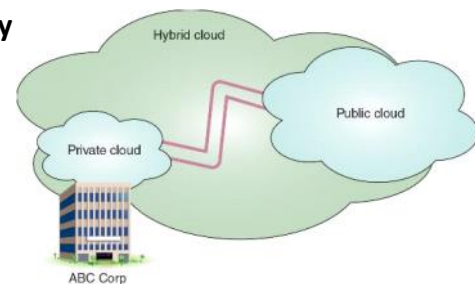


Figure 1-6 A hybrid cloud is a mix of public and private clouds or a combination of cloud and on-prem services

Community Cloud

- A **community cloud** is shared by **several organizations** that have **similar needs or goals**.
- It is **not open to the general public**.
- One organization in the group may **host and manage** the cloud, either **on-site or off-site**.
- Sometimes, a **third-party provider** manages the community cloud instead.

Community cloud = **shared by a specific group**, not public, used for **common interests** (like schools, hospitals, or government agencies).

! Cloud Within a Cloud

- **Cloud within a cloud** is a method where companies move their **existing virtual environment (vCenter)** into a **public cloud platform**.
- It allows users to keep using familiar systems while taking advantage of cloud benefits.

Advantages:

- **Cloud-native technologies** → Works well with modern cloud tools.
- **Unlimited scalability** → Can grow easily when needed.
- **Familiarity** → Users already know how to manage it.
- **Seamless migration** → Easy and smooth move to the cloud.

It means **running your private virtual system inside a public cloud**, gaining both familiarity and scalability.

☁ Common Cloud Service Models

SaaS (Software as a Service):

Software is given to users through the cloud. Example: Gmail, Google Docs, Zoom

PaaS (Platform as a Service):

A middle level of cloud that lets customers create and run applications on platforms without managing the lower infrastructure.

IaaS (Infrastructure as a Service):

Provides cloud-based network services like storage, desktops, and network devices for users to deploy and manage. Example: AWS, Microsoft Azure

SaaS = software from cloud

PaaS = build apps on cloud

IaaS = use cloud hardware

- **SaaS** includes things like email, social media, online games, and CRM systems.
- **PaaS** provides platforms for web servers and databases.
- **IaaS** offers cloud storage and virtual machines (VMs).

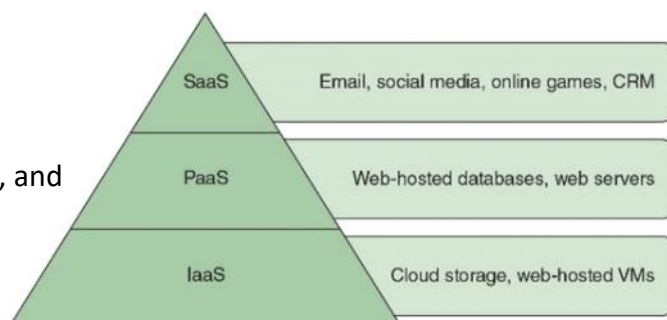


Figure 1-9 IaaS customers must understand more about configuring their cloud infrastructure—and they have more control—than do SaaS customers

! Service Model Security Concerns

- For **SaaS**, data must be encrypted both when stored and transferred. However, even encrypted data can be **exposed through social engineering attacks** that give attackers unauthorized access.
- For **PaaS**, because applications are easily accessible online, they face a **higher risk of hacking**. Providers must make sure customers don't have root or admin access that could create security weaknesses.
- For **IaaS**, **customers handle most of the security themselves** — similar to managing their own infrastructure. They must follow compliance rules, meet audit requirements, and control user identities to protect their systems.

Cloud Service Providers

Salesforce offers a popular SaaS-based CRM system.

The top three leaders in PaaS and IaaS are:

- **AWS (Amazon Web Services)**
- **Microsoft Azure**
- **GCP (Google Cloud Platform)**

Other providers include **Alibaba Cloud**, **IBM Cloud**, and **Oracle Cloud**.

You can also create your own **private cloud** using tools like **OpenStack**, **VMware**, or **Eucalyptus**.

Common Cloud Services

Major cloud providers (CSPs) offer services for setting up networks, hosting resources, and managing processes.

Common service types include:

- **Compute** – running virtual machines
- **Storage** – saving and managing data
- **Networking** – connecting systems securely
- **Security** – protecting data and access
- **Application components** – building and supporting apps
- **Management tools** – monitoring and controlling cloud systems

Internet of Things (IoT)

The **Internet of Things (IoT)** includes all smart devices connected to the Internet — such as fridges, lamps, and garage doors.

These devices use **cloud technology** to work better:

- **Communication:** IoT devices send and receive data through the Internet using cloud services.
- **Storage:** They produce large amounts of data that are often stored in the cloud for analysis and use.

Troubleshooting Methodology

A structured and careful approach to troubleshooting helps improve efficiency and avoid costly mistakes.

The **CompTIA troubleshooting method** provides a clear process to organize your thoughts and find effective solutions to problems.

Common Cloud Computing Problems

Cloud computing can face several issues, such as:

- **Connectivity issues:** no access to cloud services.
- **Latency:** slow or delayed connections.
- **Capacity:** systems can't handle too many users or data.
- **Security:** risks that may threaten data integrity.

Troubleshooting Steps

Six main steps in the CompTIA troubleshooting process:

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1. **Identify the problem** – Ask the user what happened and check for any recent changes.
 2. **Create a theory** – Think of possible reasons for the problem; sometimes it's a simple cause.
 3. **Test the theory** – Try to confirm if your idea is correct.
 4. **Make a plan and fix it** – Plan how to solve the problem and apply the fix.
 5. **Check everything** – Make sure the whole system works again and add prevention if needed.
 6. **Record the results** – Write down what you found, what you did, and what happened.

Preventative Measures

 This slide gives tips to **prevent future problems** in cloud systems:

- Use strong monitoring and analysis methods to track system health.
- Set up dashboards carefully so they show the most useful information.
- Follow proper change management rules when making updates.
- Learn how your cloud services work, where problems might happen, and how to find and fix them quickly.